

Global Renewable Energy in 2025:

How Far Have We Come?

A data-driven overview of renewable energy across primary energy, electricity generation, technologies, and regions.

~14

%

Global
primary
energy from
renewables
2024

~30

%

Global
electricity
from
renewables
2025

75% of Global GHG Emissions Still Come from Fossil Fuels

The renewable transition is also an air-quality and public-health imperative.

75%

Global greenhouse gas emissions from burning fossil fuels for energy

Energy remains the central climate challenge.

5M+

Premature deaths each year from local air pollution linked to fossil combustion

Clean power reduces CO₂ and local pollutants.

2 levers

Scale low-carbon electricity while electrifying transport and heating

Renewables and nuclear are the main low-carbon sources.



Climate forcing



Health burden



Electricity gains

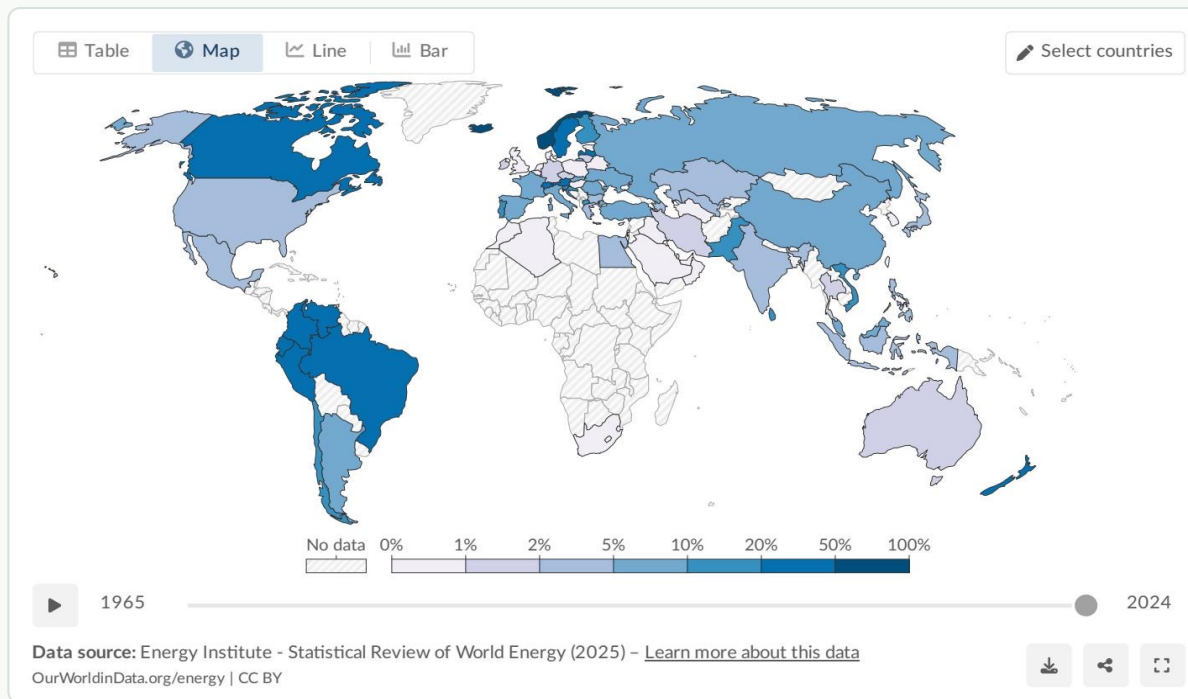


Scale-up need

Policy implication: the electricity sector is moving fastest, but total energy emissions will not fall fast enough unless transport and heating follow.

Renewables Now Supply ~14% of Global Primary Energy

Primary energy includes electricity, transport, and heating; values use the substitution method.



~14%

of global primary energy is renewable
Approximately one-seventh of all energy.

Lower than the electricity share because transport and heating remain oil- and gas-heavy. High-renewable primary-energy systems include Norway, Brazil, and New Zealand. Several fossil-fuel-producing countries in the Middle East and parts of Asia remain below 5%.

Primary energy ≠ electricity only

Renewable Electricity Generation Is Growing Rapidly, Led by Wind and Solar

Hydropower still leads, but wind and solar combined now nearly match its output.

Approx. global generation, 2024
(TWh)



Technology breakdown

Hydro	4,500 TWh	~47%
Wind	2,400 TWh	~25%
Solar	2,000 TWh	~21%
Other	700 TWh	~7%

Since ~2010, wind and solar have accelerated most sharply, reshaping the renewable mix even while hydro remains the largest source.

Almost One-Third of Global Electricity Now Comes from Renewables

Electricity is where renewables have made the most visible progress.

Share of electricity production from renewables, 2025

Renewables include solar, wind, hydropower, bioenergy, geothermal, wave, and tidal sources.

Our World
in Data

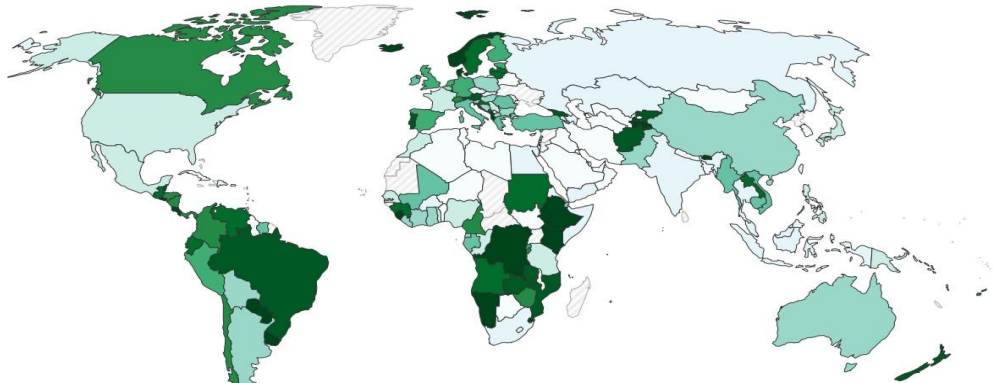
Table

Map

Line

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Select countries



1985

2025

Data source: Ember (2026); Energy Institute - Statistical Review of World Energy (2025) - [Learn more about this data](#)

OurWorldinData.org/energy | CC BY



~30%

2025

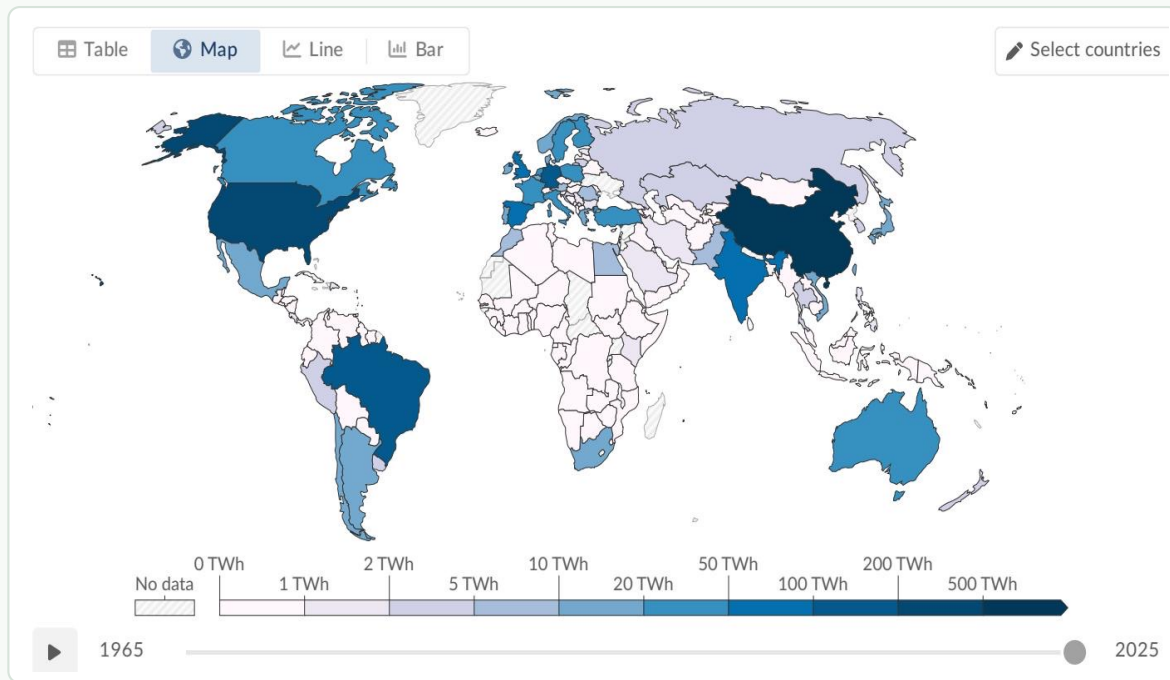
World average renewable
electricity share

Country / region	Renewable electricity
Norway	~98%
Brazil	~85%
Canada	~65%
China	~35%
India	~25%
Saudi Arabia	<2%

Regional spread: leaders reach 60–90%+ renewable electricity; many African and Middle Eastern countries remain below 10%.

Hydropower Remains the Largest Renewable Source at ~50% of Generation

A century-old technology still anchors renewable electricity systems in several regions.



~4,500 TWh

~47% of renewable electricity
Approx. global hydropower generation

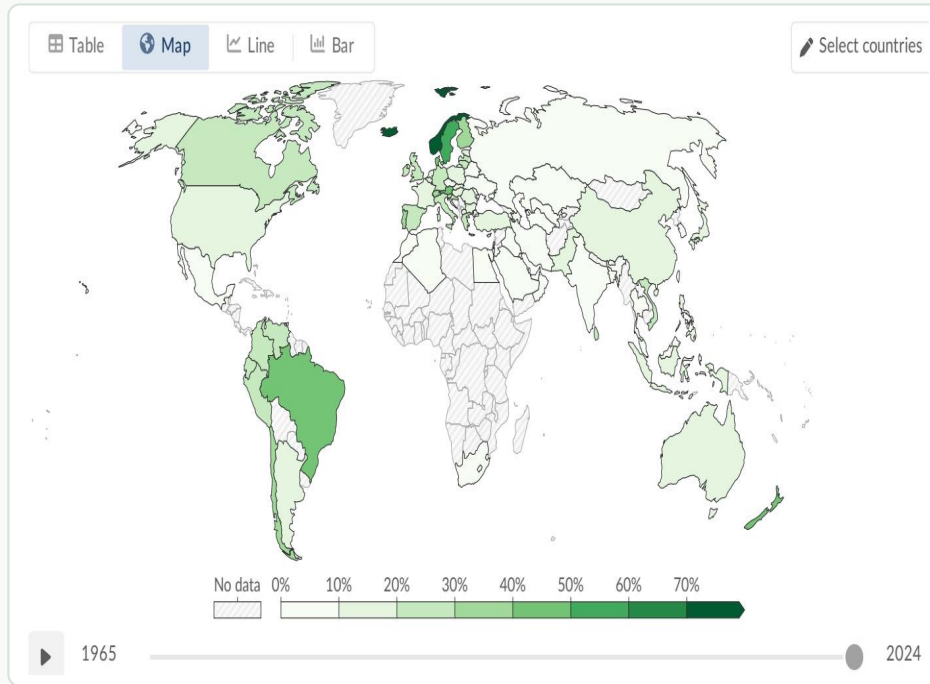
Hydropower accounts for roughly half of all renewable electricity generation. Top producers include China, Brazil, and Canada.

Growth potential is more limited than wind and solar, making diversification essential.



Wind and Solar Are the Fastest-Growing Energy Technologies in History

Use generation and renewable-electricity share data—not installed capacity—to compare their contribution.



Wind vs. solar contribution

Wind



2,400 TWh

~25% of renewable electricity

Solar



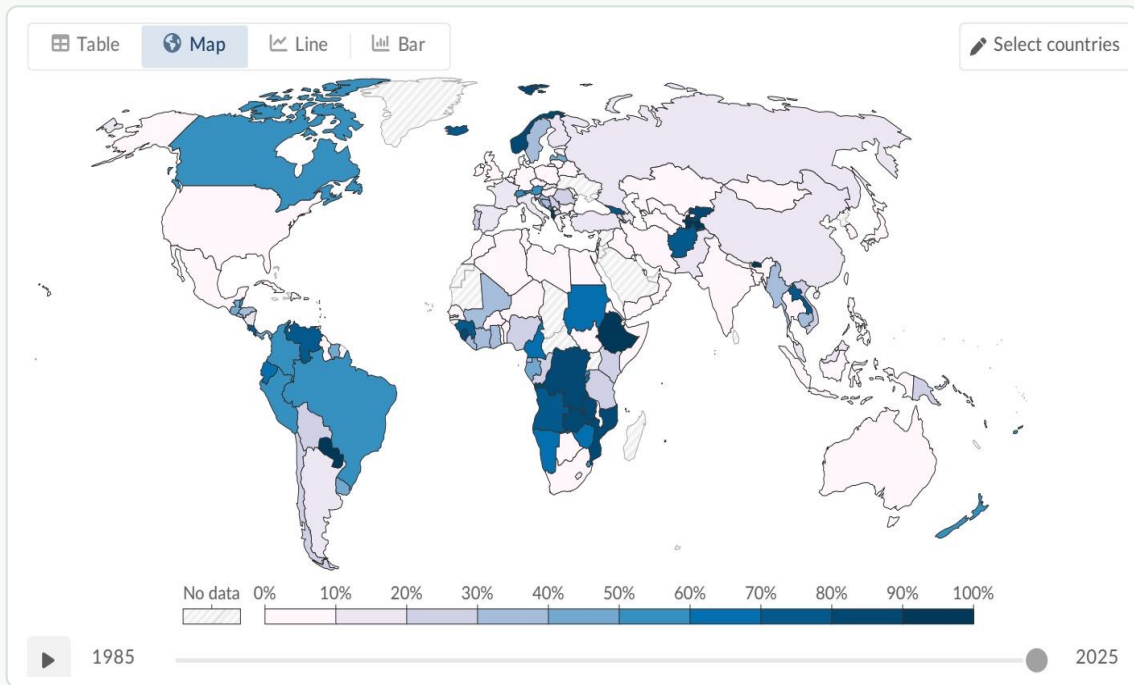
2,000 TWh

~21% of renewable electricity

Together: ~4,400 TWh — nearly matching hydropower output (~4,500 TWh). Solar is the fastest-growing source, driven by falling costs and policy support since 2010.

Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia and Latin America Lead

Resource endowments, infrastructure, and policy investment shape highly uneven renewable shares.



What explains the gap?

Scandinavia, Brazil, and Canada benefit from abundant hydropower resources. Denmark and Germany show the payoff from sustained wind and solar investment. Sub-Saharan Africa has limited renewable electricity infrastructure. Many Middle Eastern nations remain almost entirely fossil-fuel dependent.

Selected reference points: Norway ~98%, Brazil ~85%, Canada ~65%; many African and Middle Eastern countries below 10%.

Key Takeaways: Renewables Are Growing Fast but the Transition Must Accelerate

The electricity transition is real; the whole-energy transition remains incomplete.

~14%

Primary energy is renewable; transport and heating keep the total-energy share low.

~30%

Electricity is renewable globally, showing the sector where progress is most advanced.

~47%

Hydropower still provides roughly half of renewable electricity generation.

~4,400 TWh

Wind and solar together nearly match hydropower output and are growing faster.

Uneven

Regional gaps remain stark: leaders exceed 60–90%+, while many countries are below 10%.

Forward-looking call: accelerate clean electricity, grids, storage, and end-use electrification so renewable gains extend beyond the power sector.